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The Möbius Function on Arithmetic Progressions

Autor (Author):
Sabyasachi Mukherjee

Erstgutachter (First Referee):
Prof. Dr. Jörg Brüdern

Zweitgutachterin (Second Referee):
Prof. Dr. Damaris Schindler

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Notation

Throughout the text, we use the Vinogradov symbols $\ll$ and $\gg$ with their usual meaning: for two functions f and g, we write $f \ll g$ if and only if there exists a positive constant C and a real r such that $|f(x)| \leq C|g(x)|$ for all $x \geq r$. $f \gg g$ is used to mean that $g \ll f$. In addition, we also use $f = O(g)$ to mean that $f \ll g$.

Similarly, by $f = o(g)$, we mean that $\lim_{x\to\infty} \frac{f(x)}{g(x)} = 0$.

Further, (a, b) denotes the greatest common divisor of a and b. Sometimes $\gcd(a, b)$ is also used, especially when $\gcd(a, b) > 1$.

Unless otherwise mentioned, p denotes a prime.

Given a real number x, $\pi(x)$ denotes the number of prime numbers p such that $p \leq x$.

For a natural number n, $\phi(n)$ denotes the number of natural numbers $k \leq n$ such that $(k, n) = 1$.

We denote the number of distinct prime factors of n by $\omega(n)$. In particular, $\omega(1) = 0$.

The Möbius function $\mu(n)$ is defined to be $\mu(n) := (-1)^{\omega(n)}$ if n is square-free, and 0 otherwise.

Given a prime p, $v_p(n)$ denotes the highest power $k \in \mathbb{N} \cup \{0\}$ such that p^k divides n.

As usual, $e(x)$ denotes $\exp(2\pi i x)$.

We shall denote by $\#X$ the cardinality of a set X.

Given a real number x, the $\|x\|$ denotes the distance of the closest integer to x, i.e. $\|x\| = \min_{k\in\mathbb{Z}}(|k - x|)$.

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Chapter 1

Introduction

We wish to prove a Barban-Davenport-Halberstam type of theorem.
Let

$$M(N,Q) = \sum_{q \le Q} \sum_{a=1}^{q} \Bigg| \sum_{\substack{n \le N \\ n \equiv a \bmod q}} \mu(n) \Bigg|^2. \tag{1}$$

We have that

$$M(N,Q) = CQN + o(N^2), \tag{2}$$

with $C > 0$ and $1 \le Q \le N$. We wish to prove Equation 2 with an explicit estimate for the error.
Next, let c_1, c_2 and c_3 be natural numbers. Consider the expression

$$\Omega(N) = \sum_{\substack{c_1 n_1 + c_2 n_2 - c_3 n_3 = 0 \\ 1 \le n_1, n_2, n_3 \le N}} \mu(n_1)\mu(n_2)\mu(n_3). \tag{3}$$

We would like to show that $\Omega(N) = o(N^2)$. In fact, we obtain an explicit upper bound. After that, we estimate a non-trivial upper bound for

$$M_3(N,Q) = \sum_{q \le Q} \phi(q) \sum_{a=1}^{q} \Bigg(\sum_{\substack{n \le N \\ n \equiv a \bmod q}} \mu(n) \Bigg)^3. \tag{4}$$

Here we follow a strategy outlined in [7].
We note that $M(N,Q)$ and $M_3(N,Q)$ are quite similar to the objects considered in [6] and [7]. We cite two theorems by Hooley appearing in these papers, so that our calculations have more context.

Theorem 1.1 (Hooley)**.** *Suppose that*

$$\theta(x;a,k) = \sum_{\substack{p\le x\\ p\equiv a \bmod k}} \log p$$

and $E(x;a,k) = \theta(x;a,k) - \frac{x}{\phi(k)}$*. Then for* $1 \le Q \le x$ *and any positive constant* A*, we have that*

$$\sum_{k\le Q}\sum_{\substack{0<a\le k\\ (a,k)=1}} E^2(x;a,k) = Qx\log Q + DQx + O(Q^{\frac{5}{4}}x^{\frac{3}{4}}) + O\left(\frac{x^2}{(\log x)^A}\right).$$

Proof. See Theorem 1 (ii) in [6]. □

Theorem 1.2 (Hooley)**.** *Let* $E(x;a,k)$ *as before and let*

$$S(x,Q) = \sum_{k\le Q}\phi(k)\sum_{\substack{0<a\le k\\ (a,k)=1}} E^3(x;a,k).$$

Then we have

$$S(x,Q) = o(Q^{\frac{3}{2}}x^{\frac{3}{2}}\log^{\frac{3}{2}} x) + O\left(\frac{x^3}{(\log x)^A}\right)$$

as $x\to\infty$ *and* $Q = o(x/\log x)$*.*

Proof. See Theorem 1 in [7]. □

In a departure from [6], we use the circle method to compute the second Barban-Davenport-Halberstam moment of the Möbius function.

Theorem. *Suppose that* $1 \le Q \le N$*. Further, let*

$$M(N,Q) = \sum_{q\le Q}\sum_{a=1}^{q}\left|\sum_{\substack{n\le N\\ n\equiv a \bmod q}}\mu(n)\right|^2.$$

Let $C > 0$*. Then we have that*

$$M(N,Q) = \frac{1}{\zeta(2)}QN + O\left(\frac{N^2}{(\log N)^C}\right).$$

Similarly, we do a third moment calculation and aim to show that

Theorem. *Suppose that $1 \leq Q \leq N$ and let A and $A_{12} > 0$ be two constants of our choice. Then the third Barban-Davenport-Halberstam moment*

$$M_3(N,Q) = \sum_{q \leq Q} q \sum_{a=1}^{q} \left(\sum_{\substack{n \leq N \\ n \equiv a \bmod q}} \mu(n) \right)^3$$

is given by

$$M_3(N,Q) = O\left(\frac{N^3}{(\log N)^A}\right) + O\left(Q^{3/2} N^{3/2} / (\log N)^{A_{12}}\right).$$

We refer the reader to Theorem 2.7 and Theorem 4.9 which appear later in this thesis. These two theorems are somewhat analogous to the theorems cited from [6] and [7]. As far as we are aware, neither of these two computations appear in literature. The advantage of such calculations is the following: while the Riemann Hypothesis does predict square-root cancellation in the sum $\sum_{\substack{n \leq N \\ n \equiv a \bmod q}} \mu(n)$, the proof of the conjecture itself has eluded humanity for a while now. However, armed with Theorem 2.7 and Theorem 4.9, should the need to estimate either of these sums over q arise in any context, one need not assume the Riemann Hypothesis to get bounds of good quality on an average. Further, both of these calculations predict that $\sum_{\substack{n \leq N \\ n \equiv a \bmod q}} \mu(n)$ is roughly $(\frac{N}{q})^{\frac{1}{2}}$ which is in line with the $O(N^{\frac{1}{2}+\epsilon})$ bound contingent upon the Riemann Hypothesis and the conjectured $O\left(\frac{N^{\frac{1}{2}+\epsilon}}{q^{\frac{1}{2}}}\right)$.

Chapter 2

The Second Barban-Davenport-Halberstam Moment

Consider

$$M(N,Q) = \sum_{q\le Q}\sum_{a=1}^{q}\left|\sum_{\substack{n\le N\\ n\equiv a \bmod q}}\mu(n)\right|^2. \tag{5}$$

We split $M(N,Q)$ as

$$M(N,Q) = \sum_{q\le Q}\sum_{a=1}^{q}\sum_{\substack{n\le N\\ n\equiv a \bmod q}}\mu^2(n) + \sum_{q\le Q}\sum_{\substack{n\ne m\le N\\ n\equiv m \bmod q}}\mu(n)\mu(m). \tag{6}$$

We shall prove that for $1\le Q\le N$ and $C>0$ of our choice,

$$M(N,Q) = \frac{1}{\zeta(2)}QN + O\left(\frac{N^2}{(\log N)^C}\right). \tag{7}$$

We must work towards computing $\sum_{q\le Q}\sum_{\substack{n\ne m\le N\\ n\equiv m \bmod q}}\mu(n)\mu(m)$ which is equal to $2\sum_{q\le Q}\sum_{l\le N/q}\sum_{\substack{n,m\le N\\ m-n=ql}}\mu(n)\mu(m)$. This brings us to

Definition 2.1. *We define*

$$\Phi(N,Q) := \sum_{q\le Q}\sum_{l\le N/q}\sum_{\substack{n,m\le N\\ m-n=ql}}\mu(n)\mu(m), \tag{8}$$

as well as

$$S(\alpha) := \sum_{n \le N} \mu(n) e(\alpha n) \tag{9}$$

and

$$T(\alpha) := \sum_{\substack{ql \le N \\ q \le Q}} e(\alpha l q). \tag{10}$$

We note that $\Phi(N, Q) = \displaystyle\int_0^1 T(\alpha)|S(\alpha)|^2 \mathrm{d}\alpha$.

Definition 2.2. *Let $c > 1$ and $R = (\log N)^c$. Set $\mathcal{I} = [\frac{R}{N}, \frac{R}{N} + 1]$ and $\mathfrak{M}(b, r) := \{\alpha \in I : |\alpha - \frac{b}{r}| \le \frac{R}{rN}, 1 \le r \le R, (b, r) = 1$ and $1 \le b \le r\}$. The major arcs $\mathfrak{M}$ are defined as $\mathfrak{M} := \displaystyle\bigcup_{\substack{(b,r)=1 \\ 1 \le b \le r}} \mathfrak{M}(b, r)$. The rest of the interval $\mathcal{I}$, $\mathfrak{m} := \mathcal{I} \setminus \mathfrak{M}$ is termed as the minor arcs.*

Lemma 2.3. *For every $C > 0$, we have that $\sup_{\alpha \in \mathfrak{m}} |T(\alpha)| \ll \dfrac{N}{(\log N)^C}$.*

Proof. We break this proof up into two cases: $Q \le \sqrt{N}$ and $Q > \sqrt{N}$. Firstly, note that by the Dirichlet Approximation theorem (see Lemma 4.10), we can approximate $\alpha \in \mathfrak{m}$ by a rational $\frac{b}{r}$ with $(b, r) = 1, 1 \le r \le \frac{N}{R}$ such that

$$\left|\alpha - \frac{b}{r}\right| \le \frac{R}{rN} \le \frac{1}{r^2}.$$

Suppose that $Q \le \sqrt{N}$. Consider the exponential sum $T(\alpha)$ defined above. It can be estimated via the reciprocal sum lemma (see Lemma 4.13) as follows:

$$T(\alpha) = \sum_{q \le Q} \sum_{l \le \frac{N}{q}} e(\alpha l q) \ll \sum_{q \le Q} \left(\frac{N}{q}, \frac{1}{\|\alpha q\|}\right) \ll \left(\frac{N}{r} + Q + r\right) \log 2rQ, \tag{11}$$

where $\|x\|$ denotes the distance to the closest integer to x (in other words, $\|x\| = \min_{k \in \mathbb{Z}}(|k - x|)$).
We note that on minor arcs, $r > R = (\log N)^c$. For $\alpha \in \mathfrak{m}$, the first error term is (using $r > R$ and $r \le \frac{N}{R}$, $Q \le N$), $\frac{N}{r} \log 2rQ \ll \frac{N}{(\log N)^{c-1}}$. The second error term in (11) is $Q \log 2rQ \ll \sqrt{N} \log(2\frac{N}{R}\sqrt{N}) \ll \frac{N}{(\log N)^{c-1}}$, using $Q \le \sqrt{N}$. The third error term is $r \log 2rQ \le \frac{N}{R} \log 2\frac{N}{R} Q \ll \frac{N}{(\log N)^{c-1}}$.

For the case $Q > \sqrt{N}$, we split $T(\alpha)$ into three sums:

$$T(\alpha) = \left(\sum_{l \le \sqrt{N}} \sum_{\sqrt{N} < q \le \min(\frac{N}{l}, Q)} + \sum_{q \le \min(\sqrt{N}, Q)} \sum_{\sqrt{N} < l \le \frac{N}{q}} + \sum_{l \le \sqrt{N}} \sum_{q \le \min(\sqrt{N}, Q)} \right) e(\alpha l q). \quad (12)$$

The third sum is (by virtue of Lemma 4.14)

$$\sum_{q, l \le \sqrt{N}} e(\alpha l q) \ll \sum_{q \le \sqrt{N}} \left(\sqrt{N}, \frac{1}{\|\alpha q\|} \right) \ll \left(r + \sqrt{N} + \sqrt{N} + \frac{N}{r} \right) \max\{1, \log r\}. \quad (13)$$

Using $r > R$ on minor arcs, we see that the sum in (13) satisfies

$$\sum_{q, l \le \sqrt{N}} e(\alpha l q) \ll \frac{N}{(\log N)^{c-1}}.$$

The first and second sum are identical, so we shall deal only with the first one. On account of $Q > \sqrt{N}$, the second sum is

$$\sum_{q \le \sqrt{N}} \sum_{\sqrt{N} < l \le \frac{N}{q}} e(\alpha l q)$$

which is susceptible to the same treatment as the one that follows for the first sum.
We note that when $\min(\frac{N}{l}, Q) = Q$, then $N/l \ge Q$ which implies that $l \le \frac{N}{Q}$. The case that $\min(\frac{N}{l}, Q) = \frac{N}{l}$ arises when $l \ge \frac{N}{Q}$. The first sum is then rewritten as:

$$\sum_{l \le \sqrt{N}} \sum_{\sqrt{N} < q \le \min(\frac{N}{l}, Q)} e(\alpha l q) = \left(\sum_{l \le \frac{N}{Q}} \sum_{\sqrt{N} < q \le Q} + \sum_{\frac{N}{Q} < l \le \sqrt{N}} \sum_{\sqrt{N} < q \le \frac{N}{l}} \right) e(\alpha l q). \quad (14)$$

Summing up the geometric progression in q in the second sum, we get

$$\begin{aligned} \sum_{\frac{N}{Q} < l \le \sqrt{N}} \sum_{\sqrt{N} < q \le \frac{N}{l}} e(\alpha l q) &= \sum_{\frac{N}{Q} < l \le \sqrt{N}} \left(\sum_{\lfloor \sqrt{N} \rfloor + 1 \le q \le \lfloor \frac{N}{l} \rfloor} e(\alpha l)^q \right) \\ &= \sum_{\frac{N}{Q} < l \le \sqrt{N}} e(\alpha l)^{\lfloor \sqrt{N} \rfloor + 1} \left(\frac{e(\alpha l \lfloor \frac{N}{l} \rfloor - \lfloor \sqrt{N} \rfloor) - 1}{e(\alpha l) - 1} \right). \end{aligned}$$

The absolute value of this sum, by the triangle inequality, is

$$\left|\sum_{l\le\sqrt{N}}\sum_{\sqrt{N}<q\le\frac{N}{l}} e(\alpha lq)\right| \ll \sum_{l\le\sqrt{N}} \frac{1}{|e(\alpha l)-1|}$$
$$= \sum_{l\le\sqrt{N}} \frac{1}{\left|e(\frac{\alpha l}{2})-e(\frac{-\alpha l}{2})\right|} = \sum_{l\le\sqrt{N}} \frac{1}{|\sin\pi\alpha l|}$$

We note that $0 \le \|x\| \le \frac{1}{2}$ for a real number x, and that $x = k \pm \|x\|$ for some integer k. This forces $|\sin(\pi x)| = \sin(\pi\|x\|)$, using which, the above inequality transforms into

$$\left|\sum_{l\le\sqrt{N}}\sum_{\sqrt{N}<q\le\frac{N}{l}} e(\alpha lq)\right| \ll \sum_{l\le\sqrt{N}} \frac{1}{\sin(\pi\|\alpha l\|)} \ll \sum_{l\le\sqrt{N}} \frac{1}{\|\alpha l\|}, \tag{15}$$

as long as $\|\alpha l\| \ne 0, \frac{1}{2}$. Here we have used that for $0 < x < \frac{1}{2}$, the inequality $2x < \sin\pi x$ holds (see Lemma 4.6, page 103, [12] for a proof). Note that $\alpha l = 0 \implies \alpha = 0$, which is impossible as $0 \notin \mathcal{I}$, while $\alpha l = \frac{1}{2}$ forces $e(\alpha lq) = 1$ for q even and -1 for q odd. To account for this case, we estimate $e(alq)$ trivially. Therefore, by Equation 15 and our observation,

$$\sum_{\frac{N}{Q}<l\le\sqrt{N}}\sum_{\sqrt{N}<q\le\frac{N}{l}} e(\alpha lq) \ll \sum_{l\le\sqrt{N}} \min\left(\frac{N}{l}, \frac{1}{\|\alpha l\|}\right)$$
$$\ll \left(\frac{N}{r}+\sqrt{N}+r\right)\log 2r\sqrt{N} \ll \frac{N}{(\log N)^{c-1}}, \tag{16}$$

which we have obtained using $r > R$ on minor arcs and $r \le \frac{N}{R}$ for $R = (\log N)^c$.
In the same vein, the first sum in Equation 14 satisfies

$$\sum_{l\le\frac{N}{Q}}\sum_{\sqrt{N}<q\le Q} e(\alpha lq) \ll \sum_{1\le l\le\frac{N}{Q}} \min\left(Q, \frac{1}{\|\alpha l\|}\right). \tag{17}$$

By Lemma 4.14 using $r > (\log N)^c$ and $Q > \sqrt{N}$, this sum in (17) is

$$\ll (r + \frac{N}{Q} + Q + \frac{N}{Q}\cdot Q\cdot\frac{1}{r})\max\{1, \log r\} \ll \frac{N}{(\log N)^C}.$$

This shows that we can always choose a $C > 0$ such that

$$\sup_{\alpha\in\mathfrak{m}} |T(\alpha)| \ll \frac{N}{(\log N)^C}.$$

This C can be controlled by choosing $C = c-1$ and ensuring that $c-1 > 0$. □

Before we obtain the major arcs bound, we establish another lemma.

Lemma 2.4. *Consider the sum* $S(\alpha) := \sum_{n \le N} \mu(n) e(\alpha n)$ *and suppose that* $q \le (\log N)^c$, $a \in \mathbb{Z}$ *and* $\beta \in \mathbb{R}$. *Then the following holds:*

$$S\left(\frac{a}{q} + \beta\right) \ll \frac{Nq}{\exp(c'\sqrt{\log N})}(1 + N|\beta|).$$

Proof. At a rational $\frac{a}{q}$,

$$S\left(\frac{a}{q}\right) = \sum_{b=1}^{q} e\left(\frac{ab}{q}\right) \sum_{\substack{n \le N \\ n \equiv b \bmod q}} \mu(n). \tag{18}$$

Therefore,

$$\left|S\left(\frac{a}{q}\right)\right| \le \sum_{b=1}^{q} \left|e\left(\frac{ab}{q}\right) \sum_{\substack{n \le N \\ n \equiv b \bmod q}} \mu(n)\right|. \tag{19}$$

As $|e(x)| = 1$ and $q \le (\log N)^c$, applying the Siegel-Walfisz theorem (See Lemma 4.11),we get that

$$\left|S\left(\frac{a}{q}\right)\right| \ll \frac{Nq}{\exp(c'\sqrt{\log N})}. \tag{20}$$

Partial summation (see Lemma 4.12) yields

$$S\left(\frac{a}{q} + \beta\right) = e(\beta N) \sum_{n \le N} \mu(n) e\left(\frac{an}{q}\right) - 2\pi i \beta \int_1^N \left(\sum_{n \le t} \mu(n) e\left(\frac{an}{q}\right)\right) e(\beta t) \mathrm{d}t$$
$$\ll \frac{Nq}{\exp(c'\sqrt{\log N})}(1 + N|\beta|). \tag{21}$$

□

Lemma 2.5. *The major arcs bound is given by*

$$\int_{\mathfrak{M}} T(\alpha)|S(\alpha)|^2 \mathrm{d}\alpha \ll \frac{N^2}{(\log N)^{A_2}}.$$

Proof. We recall the inequality (11),

$$T(\alpha) \ll \left(\frac{N}{r} + Q + r\right) \log 2rQ.$$

Therefore, we have the following bound (as $R = (\log N)^c$ and $r \leq R$ on major arcs):

$$\begin{aligned}
&\sup_{\alpha \in \mathfrak{M}} |T(\alpha)| \ll \left(\frac{N}{r} + Q + R\right) \log 2rQ \\
&\ll (N + Q + R) \log 2RQ \ll N \log 2RQ \\
&\ll N \log R \log Q \ll N \log(\log N)^c \log N \\
&\ll N(\log N)^2. \qquad (22)
\end{aligned}$$

Further, by Lemma 2.4, as $r \leq R = (\log N)^c$ on major arcs, we obtain

$$S\left(\frac{b}{r} + \beta\right) \ll \frac{Nr}{\exp(c'\sqrt{\log N})}(1 + N\,|\beta|).$$

Using $|\beta| \leq \frac{R}{rN}$ and $r \leq R$ on major arcs, we have that

$$(1 + N|\beta|) \leq (1 + \frac{R}{r}) \ll R/r,$$

from where it follows that for $\alpha = \frac{b}{r} + \beta$ on major arcs,

$$\left|S\left(\frac{b}{r} + \beta\right)\right|^2 \ll \frac{N^2 R^2}{\exp(2c'\sqrt{\log N})}. \qquad (23)$$

The major arcs contribution, as a result of (22) and (23) is

$$\begin{aligned}
&\int_{\mathfrak{M}} T(\alpha)|S(\alpha)|^2 \mathrm{d}\alpha \qquad (24) \\
&= \sum_{r \leq R} \sum_{\substack{b=1 \\ (b,r)=1}}^{r} \int_{\frac{-R}{rN}}^{\frac{R}{rN}} T\left(\frac{b}{r} + \beta\right) \left|S\left(\frac{b}{r} + \beta\right)\right|^2 \mathrm{d}\beta \\
&\ll \frac{R^2 N^3}{\exp(2c'\sqrt{\log N})} (\log N)^2 \sum_{r \leq R} \sum_{\substack{b=1 \\ (b,r)=1}}^{r} \frac{R}{rN} \\
&\ll \frac{R^4 N^2}{\exp(2c'\sqrt{\log N})} (\log N)^2 \\
&\ll \frac{N^2}{(\log N)^C}. \qquad (25)
\end{aligned}$$

□

Lemma 2.6. *The minor arcs bound is given by*

$$\int_{\mathfrak{m}} T(\alpha)|S(\alpha)|^2 \mathrm{d}\alpha \ll \frac{N^2}{(\log N)^C}.$$

Proof. Using Lemma 2.3, we have

$$\begin{aligned}
&\int_{\mathfrak{m}} T(\alpha)|S(\alpha)|^2 \mathrm{d}\alpha \leq \sup_{\alpha\in\mathfrak{m}} |T(\alpha)| \int_{\mathcal{I}} |S(\alpha)|^2 \mathrm{d}\alpha \\
&= \sup_{\alpha\in\mathfrak{m}} |T(\alpha)| \int_0^1 |S(\alpha)|^2 \mathrm{d}\alpha \\
&\ll \frac{N}{(\log N)^C} \int_0^1 \left(\sum_{n,m\leq N} \mu(n)\mu(m) e(\alpha(n-m)) \right) \mathrm{d}\alpha \\
&= \frac{N}{(\log N)^C} \sum_{n\leq N} \mu^2(n) \ll \frac{N^2}{(\log N)^C}.
\end{aligned}$$

□

Theorem 2.7. *Suppose that* $1 \leq Q \leq N$. *Further, let*

$$M(N,Q) = \sum_{q\leq Q} \sum_{a=1}^{q} \left| \sum_{\substack{n\leq N \\ n\equiv a \bmod q}} \mu(n) \right|^2.$$

Then for every $C > 0$ *we have that*

$$M(N,Q) = \frac{1}{\zeta(2)} QN + O\left(\frac{N^2}{(\log N)^C} \right). \tag{26}$$

Proof. As noted earlier,

$$\begin{aligned}
M(N,Q) &= \sum_{q\leq Q} \sum_{a=1}^{q} \sum_{\substack{n\leq N \\ n\equiv a \bmod q}} \mu^2(n) + \sum_{q\leq Q} \sum_{\substack{n\neq m\leq N \\ n\equiv m \bmod q}} \mu(n)\mu(m) \\
&= \sum_{q\leq Q} \sum_{n\leq N} \mu^2(n) + 2 \sum_{q\leq Q} \sum_{l\leq N/q} \sum_{\substack{n,m\leq N \\ m-n=ql}} \mu(n)\mu(m).
\end{aligned}$$

The second sum, by Equation 8 is $2\Phi(N,Q)$.
Using

$$\Phi(N,Q) = \int_{\mathcal{I}} T(\alpha)|S(\alpha)|^2 \mathrm{d}\alpha = \int_{\mathfrak{M}} T(\alpha)|S(\alpha)|^2 \mathrm{d}\alpha + \int_{\mathfrak{m}} T(\alpha)|S(\alpha)|^2 \mathrm{d}\alpha$$

and Lemma 2.5 and Lemma 2.6, we get that

$$\Phi(N,Q) = O\left(\frac{N^2}{(\log N)^C}\right). \tag{27}$$

Using the standard result $\sum_{n\leq N} \mu^2(n) = \frac{N}{\zeta(2)} + O(\sqrt{N}\exp(-c'\sqrt{N}))$ (see [16] or [14], page 198, Equation 27 for a better estimate) and that $Q \leq N$, we get that the first sum is $Q\frac{N}{\zeta(2)} + O(\frac{N^2}{(\log N)^C})$. As a result of these calculations,

$$M(N,Q) = \frac{1}{\zeta(2)}QN + O\left(\frac{N^2}{(\log N)^C}\right).$$

□

Chapter 3

Computing $\Omega(N)$

Suppose that c_1, c_2 and c_3 are natural numbers. Consider the expression

$$\Omega(N) = \sum_{\substack{c_1n_1+c_2n_2-c_3n_3=0 \\ 1\leq n_1,n_2,n_3\leq N}} \mu(n_1)\mu(n_2)\mu(n_3). \tag{28}$$

$$\Omega(N) = \int_0^1 \left(\sum_{1\leq n_1\leq N} \mu(n_1)e(c_1\alpha n_1) \right) \left(\sum_{1\leq n_2\leq N} \mu(n_2)e(c_2\alpha n_2) \right) \times \left(\sum_{1\leq n_3\leq N} \mu(n_3)e(-\alpha c_3 n_3) \right) \mathrm{d}\alpha.$$

Let $f_x(\alpha) := \sum_{n\leq N} \mu(n)e(xn\alpha)$. We can therefore rewrite $\Omega(N)$ as

$$\Omega(N) = \int_0^1 f_{c_1}(\alpha)f_{c_2}(\alpha)\overline{f_{c_3}(\alpha)}\mathrm{d}\alpha. \tag{29}$$

There is a result of Davenport (see [4] or [15]) which states that for every fixed $A > 0$, the bound $\sup_{\alpha\in\mathbb{R}} \left| \sum_{n\leq N} \mu(n)e(\alpha n) \right| = O\left(\frac{N}{(\log N)^A} \right)$ holds.
As a result, the following is true because of the arithmetic mean-geometric mean inequality:

$$\begin{aligned} \Omega(N) &\leq \sup_{\alpha\in\mathbb{R}} \left| \sum_{n\leq N} \mu(n)e(\alpha n) \right| \int_0^1 |f_{c_2}(\alpha)||f_{c_3}(\alpha)|\mathrm{d}\alpha \\ &\ll \frac{N}{(\log N)^A} \left(\int_0^1 |f_{c_2}(\alpha)|^2\mathrm{d}\alpha + \int_0^1 |f_{c_3}(\alpha)|^2\mathrm{d}\alpha \right). \end{aligned}$$

As commented above, $\int_0^1 |f_{c_2}(\alpha)|^2 \mathrm{d}\alpha, \int_0^1 |f_{c_3}(\alpha)|^2 \mathrm{d}\alpha \ll N$, from where we get that

$$\Omega(N) \ll \frac{N^2}{(\log N)^A}. \tag{30}$$

Chapter 4

The Third Barban-Davenport-Halberstam Moment

We consider the third Barban-Davenport-Halberstam moment

$$M_3(N,Q) = \sum_{q\le Q}\phi(q)\sum_{a=1}^{q}\left(\sum_{\substack{n\le N\\ n\equiv a \bmod q}}\mu(n)\right)^3. \tag{31}$$

Lemma 4.1. *Choose $A>0$. Suppose that $Q \le \frac{N}{(\log N)^A}$. Then*

$$M(N,Q) \ll \frac{N^3}{(\log N)^A}.$$

Proof. Using the trivial bound $\left(\sum_{\substack{n\equiv a \bmod q\\ n\le N}}\mu(n)\right) \ll \frac{N}{q}$, choosing $C=A$ in (26),

$$\begin{aligned} M_3(N,Q) &\ll \sum_{q\le Q}\phi(q)\sum_{a=1}^{q}\left(\frac{N}{q}\right)\left(\sum_{\substack{n\equiv a \bmod q\\ n\le N}}\mu(n)\right)^2 \\ &= N\left(\frac{QN}{\zeta(2)} + O\left(\frac{N^2}{(\log N)^A}\right)\right) = \frac{QN^2}{\zeta(2)} + O\left(\frac{N^3}{(\log N)^A}\right). \end{aligned} \tag{32}$$

Applying the bound $Q \le \frac{N}{(\log N)^A}$, we are able to finish the proof. □

From now on, we shall consider the sum

$$M_3'(N,Q) = \sum_{Q'<q\le Q} \phi(q) \sum_{a=1}^{q} \left(\sum_{\substack{n\le N \\ n\equiv a \bmod q}} \mu(n) \right)^3, \tag{33}$$

where

$$Q' = \frac{N}{(\log N)^A}.$$

We obtain that

$$\begin{aligned} M_3'(N,Q) &= \sum_{Q'<q\le Q} \phi(q) \sum_{a=1}^{q} \sum_{\substack{n_1\equiv n_2\equiv n_3\equiv a \bmod q \\ n_1,n_2,n_3\le N}} \mu(n_1)\mu(n_2)\mu(n_3) \\ &= 6 \sum_{Q'<q\le Q} \phi(q) \sum_{\substack{n_1\equiv n_2\equiv n_3 \bmod q \\ n_2<n_3<n_1\le N}} \mu(n_1)\mu(n_2)\mu(n_3) \\ &+3 \sum_{Q'<q\le Q} \phi(q) \sum_{\substack{n_1\equiv n_3 \bmod q \\ n_2=n_3<n_1\le N}} \mu(n_1)\mu(n_2)\mu(n_3) \\ &+3 \sum_{Q'<q\le Q} \phi(q) \sum_{\substack{n_1\equiv n_2 \bmod q \\ n_2<n_3=n_1\le N}} \mu(n_1)\mu(n_2)\mu(n_3) + \sum_{Q'<q\le Q} \phi(q) \sum_{n\le N} \mu^3(n). \end{aligned} \tag{34}$$

We deal with each sum separately.

Lemma 4.2. *Let $A_1 > 0$. The following bound holds for $Q \le N$:*

$$\sum_{Q'<q\le Q} \phi(q) \sum_{n\le N} \mu^3(n) \ll Q^{\frac{3}{2}} N^{\frac{3}{2}} / (\log N)^{A_1}.$$

Proof. Using the prime number theorem (see Theorem 12.7, page 309 in [9]) and the observation that $\mu^3(n) = \mu(n)$ and $Q \le N$, we have, for each $A_1 > 0$,

$$\begin{aligned} &\sum_{Q'<q\le Q} q \sum_{n\le N} \mu(n) \ll Q^2 N \exp(-c\sqrt{\log N}) \\ &\ll Q^{3/2} N^{3/2} \exp(-c\sqrt{\log N}) \\ &\ll Q^{\frac{3}{2}} N^{\frac{3}{2}} / (\log N)^{A_1}. \end{aligned} \tag{35}$$

We shall choose A_1 appropriately at a later stage. □

The following lemma estimates the second and the third sums in Equation 34.

Lemma 4.3. *Suppose that $Q \leq N$. Let $A_3 > 0$. The following bound holds for both these sums:*

$$\sum_{Q'<q\leq Q} \phi(q) \sum_{\substack{n_1 \equiv n_3 \bmod q \\ n_2 = n_3 < n_1 \leq N}} \mu(n_1)\mu(n_2)\mu(n_3) \ll Q^{3/2}N^{3/2}/(\log N)^{A_3}$$

and

$$\sum_{Q'<q\leq Q} \phi(q) \sum_{\substack{n_1 \equiv n_2 \bmod q \\ n_2 < n_3 = n_1 \leq N}} \mu(n_1)\mu(n_2)\mu(n_3) \ll Q^{3/2}N^{3/2}/(\log N)^{A_3}.$$

Proof. We shall deal with one of these sums in detail. The other one requires similar treatment.
For ease of calculation, we shall instead deal with the object

$$S_1 = \sum_{q\leq Q} \frac{\phi(q)}{q} \sum_{\substack{n_1 \equiv n_2 \bmod q \\ n_2 < n_1 \leq N \\ Q'<q\leq Q}} \mu^2(n_1)\mu(n_2). \tag{36}$$

The condition $n_1 \equiv n_2 \bmod q, n_1 > n_2$ yields $l \in \mathbb{N}$ such that $n_1 - n_2 = ql$ and $l < (\log N)^A$ (because $l = \frac{n_1-n_2}{q} \leq \frac{N}{q} < (\log N)^A$ as $q > Q' = \frac{N}{(\log N)^A}$). We use the identity $\sum_{k|d} \frac{\mu(k)}{k} = \frac{\phi(d)}{d}$ to get that

$$\begin{aligned} S_1 &= \sum_{q\leq Q} \left(\sum_{k|q} \frac{\mu(k)}{k} \right) \sum_{\substack{n_1 - n_2 = ql \\ n_1 \leq N; l<(\log N)^A \\ n_2 < n_1 - lQ' \\ Q'<q\leq Q}} \mu^2(n_1)\mu(n_2) \\ &= \sum_{l<(\log N)^A} \sum_{k\leq Q} \frac{1}{k} \sum_{n_1\leq N} \mu^2(n_1) \left(\mu(k) \sum_{\substack{n_2 < n_1 - lQ' \\ n_2 \equiv n_1 \bmod lk}} \mu(n_2) \right). \end{aligned} \tag{37}$$

If we have that $lk \leq (\log N)^{A'}$ for some $A' > 0$, then we can apply the Siegel-Walfisz theorem (see Lemma 4.11) to get that the inside sum satisfies

$$\left(\mu(k) \sum_{\substack{n_2 < n_1 - lQ' \\ n_2 \equiv n_1 \bmod lk}} \mu(n_2) \right) \ll N \exp(-c'\sqrt{\log N}). \tag{38}$$

The bounds $\sum_{n_1\leq N} \mu^2(n_1) \leq N$ and $\sum_{k\leq Q} \frac{1}{k} \ll \log Q \leq \log N$ give us that $S_1 \ll N^2 \exp(-c'\sqrt{\log N})(\log N)^A \log Q \ll N^2 \exp(-c''\sqrt{\log N})$.

For $lk > (\log N)^{A'}$, we use the trivial bound and $n_1 \leq N$ to estimate the sum in Equation 37:

$$\left(\mu(k) \sum_{\substack{n_2 < n_1 - lQ' \\ n_2 \equiv n_1 \bmod lk}} \mu(n_2)\right) \ll \frac{N}{lk} \ll \frac{N}{(\log N)^{A'}}. \tag{39}$$

That tells us that

$$S_1 \ll \frac{N^2}{(\log N)^{A'}} (\log N)^A \log Q \ll \frac{N^2}{(\log N)^{A_2}}, \tag{40}$$

as long as A' is large enough.
By partial summation (see Lemma 4.12), we obtain that

$$\begin{aligned} &\sum_{Q' < q \leq Q} \phi(q) \sum_{\substack{n_1 \equiv n_3 \bmod q \\ n_2 = n_3 < n_1 \leq N}} \mu(n_1)\mu(n_2)\mu(n_3) \\ &\ll \frac{QN^2}{(\log N)^{A_2}} \ll Q^{3/2} N^{3/2} / (\log N)^{A_3} \end{aligned} \tag{41}$$

where the last inequality follows from $Q > \frac{N}{(\log N)^A}$. By choosing A' and A_2 carefully, we can control A_3.
In order to deal with the second sum

$$\sum_{Q' < q \leq Q} \phi(q) \sum_{\substack{n_1 \equiv n_2 \bmod q \\ n_2 < n_3 = n_1 \leq N}} \mu(n_1)\mu(n_2)\mu(n_3), \tag{42}$$

we consider the sum

$$\begin{aligned} S_2 &= \sum_{q \leq Q} \frac{\phi(q)}{q} \sum_{\substack{n_1 \equiv n_2 \bmod q \\ n_2 < n_1 \leq N \\ Q' < q \leq Q}} \mu(n_1)\mu(n_2)^2 \\ &= \sum_{l < (\log N)^A} \sum_{k \leq Q} \frac{1}{k} \sum_{n_2 \leq N} \mu^2(n_2) \left(\mu(k) \sum_{\substack{n_2 + lQ' < n_1 \leq N \\ n_1 \equiv n_2 \bmod lk}} \mu(n_1)\right) \end{aligned} \tag{43}$$

and prove that $S_2 \ll \frac{N^2}{(\log N)^{A_1}}$ using similar arguments as above and then use partial summation to bound the sum in (42). □

Lemma. *Suppose that $Q \leq N$. Then we have*

$$\sum_{Q' < q \leq Q} \phi(q) \sum_{\substack{n_1 \equiv n_2 \equiv n_3 \bmod q \\ n_2 < n_3 < n_1 \leq N}} \mu(n_1)\mu(n_2)\mu(n_3) \ll Q^{3/2} N^{3/2} / (\log N)^{A_{12}}. \tag{44}$$

In this proof, we follow the treatment (and the notation) in [7]. But we need to do some preparation first.
As before, we instead work with the sum

$$S_3 = \sum_{q \le Q} \frac{\phi(q)}{q} \sum_{\substack{n_1 \equiv n_2 \equiv n_3 \bmod q \\ n_2 < n_3 < n_1 \le N \\ Q' < q \le Q}} \mu(n_1)\mu(n_2)\mu(n_3) \tag{45}$$

which, using $\sum_{d|q} \frac{\mu(d)}{d} = \frac{\phi(q)}{q}$, can be rewritten as

$$S_3 = \sum_{d \le Q} \frac{\mu(d)}{d} \sum_{\substack{n_1 \equiv n_2 \equiv n_3 \bmod q \\ n_2 < n_3 < n_1 \le N \\ Q' < q \le Q \\ d|q}} \mu(n_1)\mu(n_2)\mu(n_3). \tag{46}$$

We write

$$J(d) = \sum_{\substack{n_1 \equiv n_2 \equiv n_3 \bmod q \\ n_2 < n_3 < n_1 \le N \\ Q' < q \le Q \\ d|q}} \mu(n_1)\mu(n_2)\mu(n_3). \tag{47}$$

The conditions $n_1 \equiv n_2 \equiv n_3 \bmod q$ and $n_2 < n_3 < n_1$ imply that there exist natural numbers l_1, l_2, l_3 such that

$$\begin{aligned} n_1 - n_2 &= l_3 q, \\ n_1 - n_3 &= l_2 q, \\ n_3 - n_2 &= l_1 q. \end{aligned} \tag{48}$$

Adding the last two equations in (48), we infer that

$$l_1 + l_2 = l_3. \tag{49}$$

Let $\delta = (l_1, l_2)$ so that $l_1 = \delta l_1', l_2 = \delta l_2' = \delta l_3'$ and $(l_1', l_2', l_3') = 1$.
We also note that by

$$\frac{l_1'(n_1 - n_3)}{\delta} = \frac{l_1' l_2 q}{\delta} = l_2 \left(\frac{l_1' q}{\delta} \right) = l_2' \frac{(n_3 - n_2)}{\delta}, \tag{50}$$

which, using Equation 49 simplifies to

$$0 = l_1' n_1 + l_2' n_2 - (l_1' + l_2') n_3 = l_1' n_1 + l_2' n_2 - l_3' n_3. \tag{51}$$

We further observe that $l_3 = \frac{n_1-n_2}{q} < \frac{N}{Q'} = (\log N)^A$ and $l_3' < \frac{(\log N)^A}{\delta}$. Moreover, by (50), $n_1 \equiv n_2 \equiv n_3 \bmod \delta$.
Let $\Delta = d\delta$ and $T = \delta Q' l_3$. The conditions on T that are about to appear in Equation 53 come from $n_1 - n_2 = l_3 q$, $n_1 \leq N$ and $q > \frac{N}{(\log N)^A}$.
Our sum $J(d)$ can therefore be written as

$$J(d) = \sum_{\delta<(\log N)^A} \sum_{l_3'<\frac{(\log N)^A}{\delta}} \sum_{\substack{l_3'=l_1'+l_2' \\ (l_1',l_2')=1}} P(N, Q'\delta l_3', l_1', l_2', k\delta), \tag{52}$$

where

$$P(N, T, l_1', l_2', \Delta) := \sum_{\substack{n_1\equiv n_2\equiv n_3 \bmod \Delta \\ l_1'n_1+l_2'n_2-l_3'n_3=0 \\ \frac{N}{(\log N)^A}<T<N \\ n_2+T<n_1\leq N}} \mu(n_1)\mu(n_2)\mu(n_3). \tag{53}$$

However, we work with the more general sum

$$\begin{aligned} P_1(N, t_1, t_2, l_1, l_2, \Delta) &= \sum_{\substack{n_1\equiv n_2\equiv n_3 \bmod \Delta \\ l_1'n_1+l_2'n_2-l_3'n_3=0 \\ \frac{N}{(\log N)^A}<T<N \\ t_1<n_1\leq N; n_2\leq t_2}} \mu(n_1)\mu(n_2)\mu(n_3) \\ &= \sum_{0<b\leq\Delta} P_2(N, t_1, t_2, l_1, l_2, b, \Delta), \end{aligned} \tag{54}$$

where

$$P_2(N, t_1, t_2, l_1, l_2, b, \Delta) = \sum_{\substack{n_1\equiv n_2\equiv n_3\equiv b \bmod \Delta \\ l_1'n_1+l_2'n_2-l_3'n_3=0 \\ \frac{N}{(\log N)^A}<T<N \\ t_1<n_1\leq N; n_2\leq t_2}} \mu(n_1)\mu(n_2)\mu(n_3). \tag{55}$$

Consider the three functions

$$\begin{aligned} f_1(\theta) &= \sum_{n_1\equiv b \bmod \Delta} \mu(n_1)e(l_1'n_1\theta), \\ f_2(\theta) &= \sum_{n_2\equiv b \bmod \Delta} \mu(n_2)e(l_2'n_2\theta), \\ f_3(\theta) &= \sum_{n_3\equiv b \bmod \Delta} \mu(n_3)e(-l_3'n_3\theta), \\ & t_1 < n_1 \leq N; n_2 \leq t_2 \text{ and } n_3 \leq N. \end{aligned} \tag{56}$$

We consider the general function $f(\theta)$ of which $f_1(\theta), f_2(\theta)$ and $f_3(\theta)$ are special examples.
We define

$$f(\theta) = \sum_{\substack{u<n\leq v \\ n\equiv b \bmod \Delta}} \mu(n)e(ln\theta), (0 \leq u < v \leq N, l < (\log N)^A). \tag{57}$$

The reason why we think about these three functions f_1, f_2, f_3 is that

$$\int_0^1 f_1(\theta)f_2(\theta)f_3(\theta) = P_2(N, t_1, t_2, l_1, l_2, b, \Delta). \tag{58}$$

.
We shall use the circle method to estimate $\displaystyle\int_0^1 f_1(\theta)f_2(\theta)f_3(\theta)\mathrm{d}\theta$. To that end, we define our major arcs as follows:

Definition 4.4. *Let* $M = \frac{N}{(\log N)^A}$. *Let* $I = [\frac{(\log N)^A}{N}, \frac{(\log N)^A}{N} + 1]$ *and let* $\mathfrak{M}(h,k) := \{\theta \in I : \left|\theta - \frac{h}{k}\right| \leq \frac{1}{M}, 1 \leq k \leq (\log N)^A, (h,k) = 1$ *and* $1 \leq h \leq k\}$. *The major arcs* $\mathfrak{M}$ *are defined as* $\mathfrak{M} := \displaystyle\bigcup_{\substack{(h,k)=1 \\ 1\leq h\leq k}} \mathfrak{M}(h,k)$. *The rest of the interval* I, $m := I \setminus \mathfrak{M}$ *is called the minor arcs.*

We first talk about the minor arcs bound.

Lemma 4.5. *Suppose that* $A_4 > 0$. *We have*

$$\int_{\mathfrak{m}} f_1(\theta)f_2(\theta)f_3(\theta)\mathrm{d}\theta \ll \frac{N^2}{(\log N)^{A_4}}.$$

Proof. We have, using the triangle and the arithmetic mean-geometric mean inequalities,

$$\begin{aligned}\left|\int_{\mathfrak{m}} f_1(\theta)f_2(\theta)f_3(\theta)\mathrm{d}\theta\right| &\leq \sup_{\theta\in\mathfrak{m}} |f_1(\theta)| \int_0^1 |f_2(\theta)f_3(\theta)|\,\mathrm{d}\theta \\ &\ll \sup_{\theta\in\mathfrak{m}} |f_1(\theta)| \left(\int_0^1 |f_2(\theta)|^2\mathrm{d}\theta + \int_0^1 |f_3(\theta)|^2\mathrm{d}\theta\right).\end{aligned} \tag{59}$$

We get using the bound of Davenport (see [4] or [15])

$$f(\theta) = \frac{1}{\Delta}\sum_{0<c\leq\Delta} e\left(-\frac{bc}{\Delta}\right) \sum_{u<n\leq v} \mu(n)e\left((l\theta + \frac{c}{\Delta})n\right) \ll \frac{N}{(\log N)^{A_4}}. \tag{60}$$

Further, by the trivial bound,

$$\int_0^1 |f_2(\theta)|^2 \mathrm{d}\theta, \int_0^1 |f_3(\theta)|^2 \mathrm{d}\theta \ll N. \tag{61}$$

By (59), (60) and (61), we get that

$$\int_{\mathfrak{m}} f_1(\theta) f_2(\theta) f_3(\theta) \mathrm{d}\theta \ll \frac{N^2}{(\log N)^{A_4}}.$$

We shall choose A_4 later. □

Lemma 4.6. *Suppose that $\Delta \leq (\log N)^{A_5}$. On major arcs, the following bound holds:*

$$\int_{\mathfrak{M}} f_1(\theta) f_2(\theta) f_3(\theta) \mathrm{d}\theta \ll \frac{N^2}{(\log N)^{A_8}}$$

Proof. Consider a θ in a major arc centered at a rational $\frac{h}{k}$ and suppose that $\Delta \leq (\log N)^{A_5}$. We recall that by our definition of major arcs, $k \leq (\log N)^A$. Let θ be $\theta = \frac{h}{k} + \phi$ and $k' = k/(l,k)$ and $l' = l/(l,k)$. We expand $f(\frac{h}{k})$ as follows:

$$\begin{aligned} f\left(\frac{h}{k}\right) &= \sum_{\substack{u<n\leq v \\ n\equiv b \bmod \Delta}} \mu(n) e(hl'n/k') \\ &= \sum_{0<a'\leq k'} e(hl'a'/k') \sum_{\substack{u<n\leq w \\ n\equiv a' \bmod k' \\ n\equiv b \bmod \Delta}} \mu(n). \end{aligned} \tag{62}$$

When $\Delta \leq (\log N)^{A_5}$,we have that $[k,\Delta] \leq (\log N)^{A_5}$, and hence, by the Siegel-Walfisz theorem (see Lemma 4.11), we have that for each $A_6 > 0$, the term in Equation 62 satisfies

$$\begin{aligned} &\sum_{0<a'\leq k'} e(hl'a'/k') \sum_{\substack{u<n\leq w \\ n\equiv a' \bmod k' \\ n\equiv b \bmod \Delta}} \mu(n) \ll \phi(k') N \exp(-c'\sqrt{\log N}) \\ &\ll \frac{N}{(\log N)^{A_6}}. \end{aligned} \tag{63}$$

By (63), we obtain that

$$f\left(\frac{h}{k}\right) \ll \frac{N}{(\log N)^{A_6}}. \tag{64}$$

Finally, by partial summation and the bounds $|l'| \ll (\log N)^A$, $v \leq N$ and the information from the major arcs $|\phi| \leq (\log N)^A/N$, we have that

$$\begin{aligned} f\left(\frac{h}{k}+\phi\right) &= e(vl'\phi) f\left(\frac{h}{k}\right) \\ &\quad + O\left(|l'\phi| \int_u^v \left(\sum_{\substack{u<n\leq t \\ n\equiv b \bmod \Delta}} \mu(n) e(hl'n/k')\right) e(l'\phi t) \mathrm{d}t\right) \\ &\ll \frac{N}{(\log N)^{A_6}} + \frac{N}{(\log N)^{A_6}} v|l'\phi| \\ &\ll \frac{N}{(\log N)^{A_7}}. \end{aligned} \tag{65}$$

Integrating over major arcs and using (65), we get

$$\begin{aligned} &\int_{\mathfrak{M}} f_1(\theta) f_2(\theta) f_3(\theta) \mathrm{d}\theta \\ &= \sum_{k\leq(\log N)^{A_5}} \sum_{\substack{h=1 \\ (h,k)=1}}^{k} \int_{\frac{-(\log N)^A}{N}}^{\frac{(\log N)^A}{N}} f_1\left(\frac{h}{k}+\phi\right) f_2\left(\frac{h}{k}+\phi\right) f_3\left(\frac{h}{k}+\phi\right) \mathrm{d}\phi \\ &\ll \frac{N^3}{(\log N)^{3A_7}} \cdot \frac{(\log N)^A}{N} \sum_{k\leq(\log N)^A} \phi(k) \\ &\ll \frac{N^2}{(\log N)^{A_8}}. \end{aligned} \tag{66}$$

This completes our proof. □

Lemma 4.7. *We have that*

$$P(N, T, l_1', l_2', \Delta) \ll \frac{N^2}{(\log N)^{A_9}}. \tag{67}$$

Proof. By Equation 58, Lemma 4.5 and Lemma 4.6, choosing $A_4 = A_8$ (A_4 appears in the minor arcs argument) we get that for $\Delta \leq (\log N)^{A_5}$,

$$P_2(N, t_1, t_2, l_1, l_2, b, \Delta) \ll \frac{N^2}{(\log N)^{A_8}}. \tag{68}$$

It follows from Equation 55 and (68) that for $\Delta \leq (\log N)^{A_5}$,

$$P_1(N, t_1, t_2, l_1, l_2, \Delta) = \sum_{0<b\leq\Delta} P_2(N, t_1, t_2, l_1, l_2, b, \Delta) \ll \frac{N^2}{(\log N)^{A_9}}. \tag{69}$$

From there, it emerges that

$$P(N,T,l_1',l_2',\Delta) \ll \frac{N^2}{(\log N)^{A_9}}.$$

In the case where $\Delta > (\log N)^{A_5}$ holds, we trivially bound μ in $P(N,T,l_1',l_2',\Delta)$ to get that

$$P(N,T,l_1',l_2',\Delta) \ll \sum_{\substack{n_1\equiv n_2\equiv n_3 \bmod \Delta \\ l_1'n_1+l_2'n_2-l_3'n_3=0}} 1. \tag{70}$$

We note that once we fix $l_1', l_2', l_3', n_1, n_2$, the remaining n_3 gets fixed. Further, each n_1, n_2 can be chosen to be congruent to each other in $\frac{N}{\Delta}$ ways and the sum runs over at most Δ residue classes, and hence

$$P(N,T,l_1',l_2',\Delta) \ll \left(\frac{N}{\Delta}\right)^2 \cdot \Delta \ll \frac{N^2}{(\log N)^{A_5}}. \tag{71}$$

We can set $A_5 = A_9$. □

Lemma 4.8. *The sum* $S_3 = \sum_{q\le Q} \frac{\phi(q)}{q} \sum_{\substack{n_1\equiv n_2\equiv n_3 \bmod q \\ n_2<n_3<n_1\le N \\ Q'<q\le Q}} \mu(n_1)\mu(n_2)\mu(n_3)$ *satisfies*

$$S_3 \ll \frac{N^2}{(\log N)^{A_{11}}}.$$

Proof. We remind ourselves from Equations 46, 47 and 52 that we are trying to evaluate

$$\begin{aligned} S_3 &= \sum_{d\le Q} \frac{\mu(d)}{d} \sum_{\substack{n_1\equiv n_2\equiv n_3 \bmod q \\ n_2<n_3<n_1\le N \\ d|q \\ Q'<q\le Q}} \mu(n_1)\mu(n_2)\mu(n_3) \\ &= \sum_{d\le Q} \frac{\mu(d)}{d} \sum_{\delta<(\log N)^A} \sum_{l_3'<\frac{(\log N)^A}{\delta}} \sum_{\substack{l_3'=l_1'+l_2' \\ (l_1',l_2')=1}} P(N,Q'\delta l_3',l_1',l_2',d\delta) \end{aligned} \tag{72}$$

We first sum over d to obtain that, using Lemma 4.7 and $\log Q \le \log N$,

$$\sum_{d\le Q} \frac{\mu(d)}{d} P(N,Q'\delta l_3',l_1',l_2',d\delta) \ll \frac{N^2}{(\log N)^{A_9}}(\log Q) \ll \frac{N^2}{(\log N)^{A_{10}}}. \tag{73}$$

As a result,

$$\sum_{\delta<(\log N)^A} \sum_{l_3'<\frac{(\log N)^A}{\delta}} \sum_{\substack{l_3'=l_1'+l_2' \\ (l_1',l_2')=1}} \sum_{k\leq Q} \frac{\mu(k)}{k} P(N, Q'\delta l_3', l_1', l_2', d\delta) \ll \frac{N^2}{(\log N)^{A_{11}}} \tag{74}$$

which is what we wanted to show. In the last inequality, we have utilised that $l_2', l_1' < (\log N)^{A_5}/\delta$. □

Theorem 4.9. *Suppose that $1 \leq Q \leq N$. Let $A > 0$ and $A_{12} > 0$. Then the third Barban-Davenport-Halberstam moment*

$$M_3(N,Q) = \sum_{q\leq Q} \phi(q) \sum_{a=1}^{q} \left(\sum_{\substack{n\leq N \\ n\equiv a \bmod q}} \mu(n) \right)^3 \tag{75}$$

is given by

$$M_3(N,Q) = O\left(\frac{N^3}{(\log N)^A}\right) + O\left(Q^{3/2}N^{3/2}/(\log N)^{A_{12}}\right). \tag{76}$$

Proof. We have shown in Lemma 4.8 that

$$\sum_{d\leq Q} \frac{\phi(d)}{d} \sum_{\substack{n_1\equiv n_2\equiv n_3 \bmod q \\ n_2<n_3<n_1\leq N \\ d|q \\ Q'<d\leq Q}} \mu(n_1)\mu(n_2)\mu(n_3) \ll \frac{N^2}{(\log N)^{A_{11}}}.$$

By this result and partial summation, we infer that

$$\sum_{Q'<q\leq Q} \phi(q) \sum_{\substack{n_1\equiv n_2\equiv n_3 \bmod q \\ n_2<n_3<n_1\leq N}} \mu(n_1)\mu(n_2)\mu(n_3) \ll \frac{QN^2}{(\log N)^{A_{11}}}. \tag{77}$$

Making use of $Q > \frac{N}{(\log N)^{A_5}}$, we finally get that

$$\sum_{Q'<q\leq Q} \phi(q) \sum_{\substack{n_1\equiv n_2\equiv n_3 \bmod q \\ n_2<n_3<n_1\leq N}} \mu(n_1)\mu(n_2)\mu(n_3) \ll \frac{Q^{\frac{3}{2}}N^{\frac{3}{2}}}{(\log N)^{A_{12}}}. \tag{78}$$

Finally, putting together Equation 31, Lemma 4.1, Lemma 4.2 and Lemma 4.3, Equations 33 and 34, and (78) and choosing $A_1 = A_3 = A_{12}$, we get that

$$M_3(N,Q) = O\left(\frac{N^3}{(\log N)^A}\right) + O\left(Q^{3/2}N^{3/2}/(\log N)^{A_{12}}\right). \tag{79}$$

□

Appendix

In this section, we state we state without proof some results which we have made use of.

Lemma 4.10 (Dirichlet's Approximation Theorem)**.** *Suppose that α is a real number. Then for every real $X \geq 1$, there exists a rational $\frac{a}{q}$ such that $(a, q) = 1, 1 \leq q \leq X$ and $\left|\alpha - \frac{a}{q}\right| \leq \frac{1}{qX}$.*

Proof. See Lemma 2.1, page 9 in [13]. □

Lemma 4.11 (Corollary to the Siegel-Walfisz Theorem)**.** *Suppose that A is a positive real and $q \leq (\log x)^A$, then there exists a constant c depending on A such that*

$$\sum_{\substack{n \leq x \\ n \equiv a \bmod q}} \mu(n) \ll x \exp(-c\sqrt{\log x})$$

for all $a \in \mathbb{Z}$.

Proof. See Exercise 13, page 385 in [11]. □

Lemma 4.12 (Partial Summation Formula)**.** *Suppose that $f : [a, b] \to \mathbb{C}$ is a C^1-function i.e. it is continuously differentiable and r is a natural number. Further (a_n) is a sequence of complex numbers and we denote by $A(x)$ the sum*

$$A(x) = \sum_{r \leq n \leq x} a_n.$$

Then we have

$$\sum_{r \leq n \leq x} a_n f(n) = A(x)f(x) - \int_r^x A(u)f'(u)\mathrm{d}u.$$

Proof. See Lemma 1.1.3 on page 9 in [1]. □

Lemma 4.13 (Reciprocal sum lemma)**.** *Suppose that α is a real number. If*

$$\left|\alpha - \frac{a}{q}\right| \leq \frac{1}{q^2},$$

where $q \geq 1$ and $(a, q) = 1$, then for any positive integer n and any real number $U \geq 1$, the following holds:

$$\sum_{1 \leq k \leq U} \min\left(\frac{n}{k}, \frac{1}{\|\alpha k\|}\right) \ll \left(\frac{n}{q} + U + q\right) \log 2qU.$$

Proof. See Lemma 4.10, page 108 in [12]. □

Lemma 4.14. *Let α be a real number. If we have the approximation*

$$\left|\alpha - \frac{a}{q}\right| \leq \frac{1}{q^2},$$

with $q \geq 1$ and $(a, q) = 1$, then for any reals U and n, we have

$$\sum_{1 \leq k \leq U} \min\left(n, \frac{1}{\|\alpha k\|}\right) \ll \left(q + U + n + \frac{Un}{q}\right) \max\{1, \log q\}.$$

Proof. See Lemma 4.11, page 110 in [12]. □

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[15] Error Term in Davenport's sum $\sum_{n\leq x}\mu(n)e(2\pi nx)$ (Mathoverflow thread).

[16] Show that $\sum_{n\leq x}\mu^2(n) = \frac{x}{\zeta(2)} + o(\sqrt{x})$ $(x \to \infty)$ (Math.Stackexchange thread).